SAT Structural Hardness II

Graph-Local Defects and Structural Hardness in Random 3-SAT

Christof Krieg

MAAT Project – Independent Research

https://maat-research.com

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Abstract

Paper 49 identified SAT and constraint systems as the main Achilles test for the universality of MAAT-style structural selection. The previous SAT validation used coarse global defect fields and performed poorly, indicating that the available feature basis was not yet adequate for combinatorial hardness. This paper addresses that weakness by constructing a new graph-local SAT benchmark.

We generate a deterministic ensemble of 520 random 3-SAT instances with $24 \leq n \leq 36$ variables and clause-density range $3.17 \leq \alpha = m/n \leq 5.22$. For each instance we compute graph-local and propagation-local diagnostics: constraint-graph expansion, clause-variable interaction cycles, backdoor-density proxy, local covariance conditioning, unit-propagation depth, contradiction rate, literal imbalance, and v1.2.1-style emergent robustness. The target is a bounded DPLL node-count proxy,

$$y = \log(1 + \text{DPLL nodes}).$$

The result is mixed but informative. Scalar F_{MAAT} and stability-only scores fail on this SAT benchmark. Graph-local MAAT diagnostics beat shuffled-null structure and improve over density-only when combined with the SAT phase-density channel:

$$R^2_{\text{MAAT+density}} = 0.2347 \quad \text{vs.} \quad R^2_{\text{density}} = 0.1886.$$

The primary quantitative result is therefore the gain

$$\Delta R^2 = +0.0461,$$

indicating that the graph-local structural feature map retains nontrivial hardness information beyond phase-transition proximity alone. However, a simple standard graph baseline remains slightly stronger,

$$R^2_{\text{graph}} = 0.2469.$$

Thus Paper 50 does not close the SAT universality problem. It narrows it: SAT hardness requires graph-local structural defects, but the current MAAT compression is not yet superior to ordinary graph features.

1 Purpose and Status

The purpose of this paper is to answer the critique raised by Paper 49: if MAAT is proposed as a universal structural-selection framework, SAT should not remain a weak point. SAT is a strong test because it is objective, combinatorial, and far from the field-theory and cosmology settings where structural selection first appeared.

The present work is not a proof of $P \neq NP$, not a SAT solver competition, and not a claim that MAAT now explains computational hardness. It is a controlled benchmark asking a narrower question:

Do graph-local MAAT-style defects improve SAT hardness prediction beyond density-only, stability-only, scalar MAAT, and shuffled-null baselines?

The graph-local diagnostics are solver-independent inputs, but the target is a specific deterministic DPLL node-count proxy. Full solver-independent hardness validation therefore remains a future cross-solver benchmark rather than a claim of the present paper.

2 Synthetic SAT Ensemble

The benchmark generates random 3-SAT formulas with fixed seed 50050. For each formula:

- the variable count is sampled in the range $24 \leq n \leq 36$;
- the clause density spans $3.17 \leq \alpha = m/n \leq 5.22$;
- clauses contain three distinct variables with random signs;
- a deterministic DPLL solver with unit propagation and a degree-based branching heuristic records the node count.

The dataset contains 520 instances. No run reaches the DPLL node cap. The satisfiable fraction is

$$0.6346.$$

3 Graph-Local Defects

The benchmark adds diagnostics that were missing from the earlier SAT attempt. They are not derived from the DPLL trace, except for the target node count. The structural inputs are:

- **constraint-graph expansion:** sampled variable subsets are scored by the number of clauses they touch;
- **clause-variable interaction cycles:** triangle density in the variable co-occurrence graph acts as a short-cycle proxy;
- **backdoor-density proxy:** a greedy hitting set estimates the fraction of variables needed to touch 80% of clauses;
- **local covariance conditioning:** variable-neighborhood feature covariances define a conditioning score;
- **unit-propagation depth:** single-literal probes measure how much forced propagation is induced;
- **contradiction rate:** single-literal probes measure immediate contradiction frequency;
- **literal and degree balance:** polarity imbalance and degree dispersion measure local balance.

These quantities are mapped to supports H, B, S, V and emergent robustness:

$$R_{\text{resp}} = (HBV)^{1/3}, \qquad R_{\text{rob}} = \min\Big(R_{\text{resp}}, (HBSV)^{1/4}\Big). \tag{1}$$

The scalar structural score is

$$F_{\text{MAAT}} = -\log H - \log B - \log S - \log V - \log R_{\text{rob}}, \tag{2}$$

with a small numerical ϵ in the implementation.

4 Baselines

Five predictors are compared by 5-fold cross-validation on $\log(1 + \text{DPLL nodes})$:

- **density-only:** α and a phase-transition proximity score;
- **standard graph:** density, degree variation, maximum degree, literal imbalance, and interaction cycles;
- **stability-only:** R_{rob} alone;
- **MAAT scalar:** F_{MAAT} alone;
- **MAAT graph-local plus density:** the full graph-local MAAT feature set plus α and phase-density proximity.

A shuffled-null test permutes the target and recomputes the MAAT graph-local Spearman distribution.

5 Results

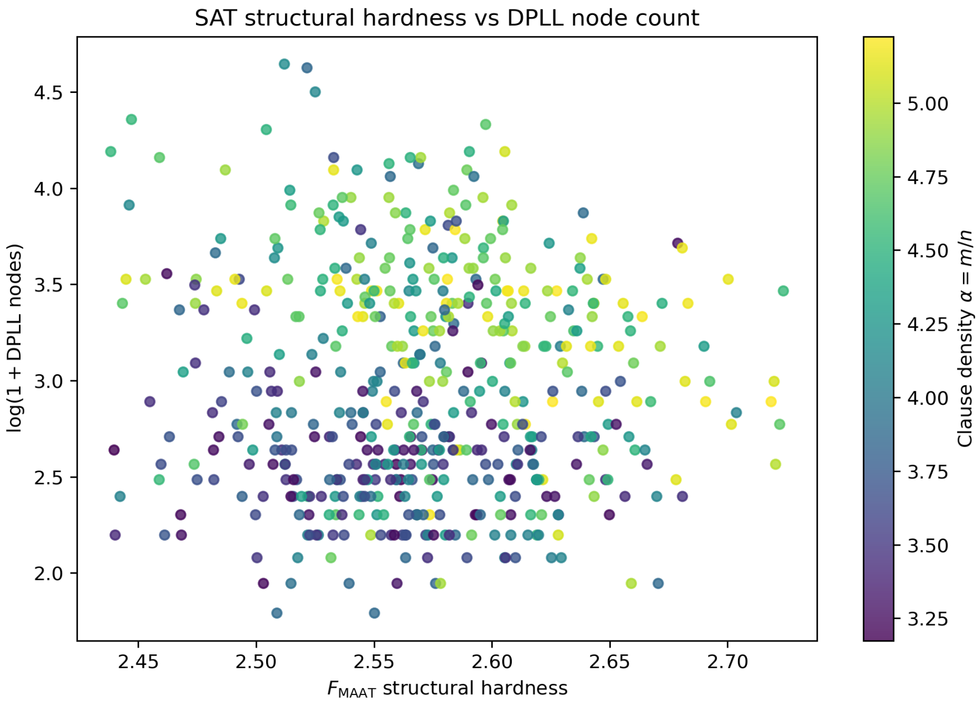

Figure 1: Scalar F_{MAAT} versus DPLL node count. The scalar compressed score alone is not sufficient for SAT hardness.

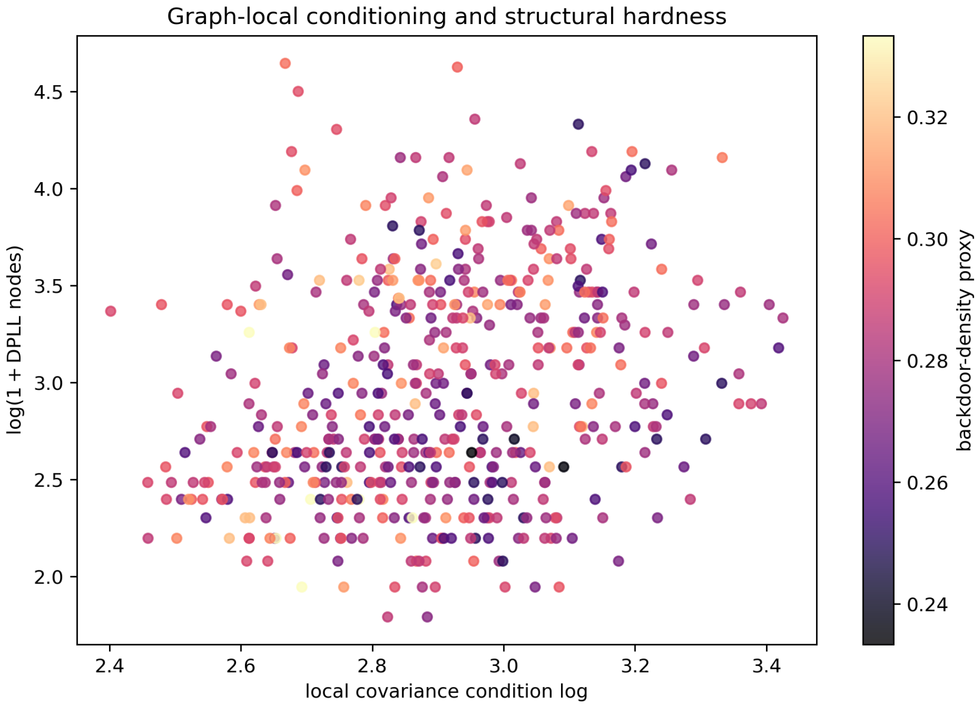

Figure 2: Local covariance conditioning versus DPLL node count. Conditioning carries visible hardness signal and is one of the more useful graph-local diagnostics.

Table 1: Five-fold cross-validated hardness prediction.

Model	$\mathbf{R^2}$	**Pearson**	**Spearman**	**RMSE**
Density-only	0.1886	0.4363	0.4617	0.5174
Standard graph	**0.2469**	**0.4992**	**0.5162**	**0.4985**
Stability-only	−0.0145	−0.1145	−0.1338	0.5786
MAAT scalar	−0.0145	−0.1152	−0.1345	0.5786
MAAT graph-local	0.1469	0.3848	0.3914	0.5306
MAAT graph-local+density	0.2347	0.4897	0.5088	0.5025

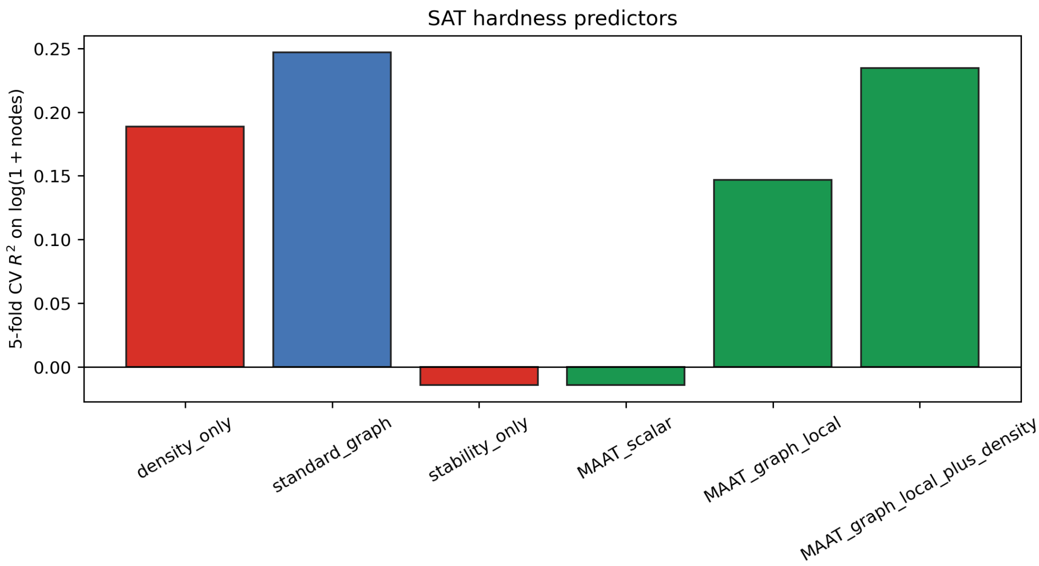

Figure 3: Model comparison. The graph-local MAAT+density model improves over density-only but remains slightly below the standard graph baseline.

The scalar correlations are:

Score	Pearson	Spearman
F_{MAAT}	-0.0296	0.0029
Density peak	0.0909	0.0833
$1 - R_{\mathrm{rob}}$	-0.0299	0.0029
Covariance conditioning	0.2752	0.2918
Backdoor-density proxy	0.1215	0.1257

The shuffled-null 95th percentile for MAAT graph-local Spearman correlation is 0.0623. The observed MAAT graph-local Spearman value is 0.3914, so the structural feature map is not a shuffled-null artifact.

6 Interpretation

The result is a partial success and a useful failure. The partial success is that graph-local structural features clearly contain SAT hardness information:

$$\rho_S = 0.3914$$

for MAAT graph-local features, well above the shuffled-null 95th percentile. Adding phase-density information gives

$$R^2 = 0.2347,$$

which improves over density-only by

$$\Delta R^2 = 0.0461.$$

The failure is equally important. The standard graph baseline remains slightly stronger:

$$R^2_{\mathrm{graph}} - R^2_{\mathrm{MAAT+density}} = 0.0122.$$

Moreover, scalar F_{MAAT} and stability-only scores fail almost completely. This shows that SAT hardness cannot be captured by a single global robustness number or a compressed support score. In other words, the scalarization

$$d_a \to \Gamma_a \to F_{\mathrm{MAAT}}$$

is too lossy for the present SAT ensemble. The useful signal survives at the level of local defect fields, especially conditioning, propagation, cycle, and backdoor-density structure. This suggests a multi-scale frustration interpretation:

$$\text{Hardness} \not\sim f(\text{global robustness}), \qquad \text{Hardness} \sim f(\text{multi-scale frustration geometry}). \tag{3}$$

The main lesson is:

> *SAT hardness appears to require graph-local structural diagnostics. MAAT-style compression helps only when the phase-density channel is retained, and it is not yet superior to ordinary graph features.*

7 What Changed Relative to the Earlier SAT Failure

The earlier SAT validation performed poorly with global fields. Paper 50 improves the situation in three ways:

- it replaces global summaries by graph-local expansion, conditioning, cycle, and backdoor-density proxies;
- it compares against explicit density-only, graph, stability-only, scalar-MAAT, and shuffled-null baselines;
- it reports both improvement and remaining failure rather than presenting SAT as solved.

The most useful individual scalar is local covariance conditioning, with Spearman correlation

$$\rho_S = 0.2918.$$

This supports the hypothesis from earlier SAT notes that hardness is partly geometric: badly conditioned local constraint neighborhoods correlate with increased solver effort.

8 Limitations

Several limitations remain:

- the formulas are synthetic random 3-SAT instances, not industrial or competition benchmarks;
- the DPLL target depends on a particular deterministic branching heuristic;
- the backdoor measure is a greedy proxy, not an exact strong-backdoor computation;
- no CDCL solver statistics are included;
- no independent SAT dataset is used;
- the graph-local MAAT compression still underperforms the standard graph baseline by a small margin.

9 Reproducibility

Repository:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/sat_structural_hardness_paper50

Run:

```
python3 sat_structural_hardness_paper50.py
```

The script writes:

- `outputs_paper50/paper50_sat_instances.csv`;
- `outputs_paper50/paper50_scalar_correlations.csv`;
- `outputs_paper50/paper50_model_comparison.csv`;
- `outputs_paper50/paper50_summary.json`;
- `outputs_paper50/fig1_fmaat_vs_dpll_nodes.png`;
- `outputs_paper50/fig2_conditioning_vs_nodes.png`;
- `outputs_paper50/fig3_model_comparison.png`.

The benchmark generates synthetic SAT formulas with a fixed random seed. No external SAT dataset is redistributed.

10 Conclusion

Paper 50 directly addresses the SAT weakness identified in Paper 49. It does not solve it, but it sharpens it. Graph-local structural diagnostics improve over density-only and strongly beat shuffled-null structure, but the compressed MAAT representation is not yet superior to an ordinary graph-feature baseline.

This is a scientifically useful result. It shows that SAT hardness is not captured by global stability alone and that future MAAT-SAT work must remain graph-local, phase-aware, and benchmarked against strong ordinary graph baselines. The next SAT step should use external SAT libraries or benchmark instances, CDCL statistics, exact or stronger backdoor proxies, and predeclared cross-instance validation.

References

[1] S. A. Cook, "The complexity of theorem-proving procedures," *Proceedings of the Third Annual ACM Symposium on Theory of Computing*, 151–158 (1971).

[2] D. Mitchell, B. Selman, and H. Levesque, "Hard and easy distributions of SAT problems," *Proceedings of AAAI* (1992).

[3] R. Monasson, R. Zecchina, S. Kirkpatrick, B. Selman, and L. Troyansky, "Determining computational complexity from characteristic phase transitions," *Nature* **400**, 133–137 (1999). DOI: 10.1038/22055.

[4] R. Williams, C. P. Gomes, and B. Selman, "Backdoors to typical case complexity," *Proceedings of IJCAI* (2003).

[5] C. Krieg, "Universal Structural Selection Benchmarks," preprint (2026).